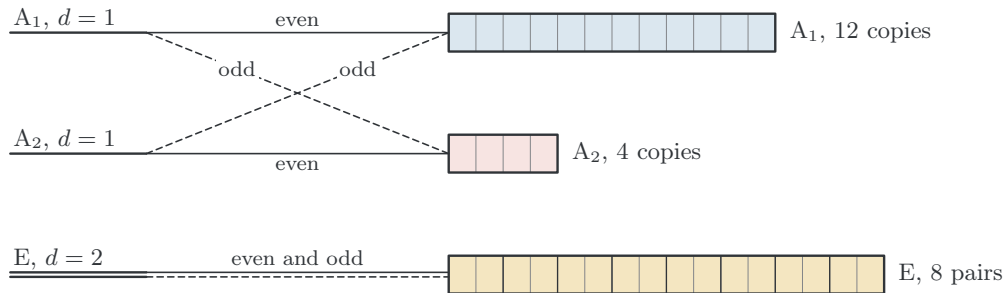


$\mathcal{H}_{\text{spin}} = (\mathbb{C}^2)^{\otimes 5}$  for the five protons,  $\dim = 32$ ; under  $G_6$  it holds 12 copies of  $A_1$ , 4 of  $A_2$  and 8 of E, so  $12 + 4 + 2 \cdot 8 = 32$ ; at right each spin species is a bar of one unit per dimension

spatial species  $\Gamma$ , dimension  $d_\Gamma$

spin species, copies in  $\mathcal{H}_{\text{spin}}$



the weight of a spatial level is the length of the bar it is joined to:  $w_{A_1}(+) = 12$ ,  $w_{A_1}(-) = 4$ ;  $w_{A_2}(+) = 4$ ,  $w_{A_2}(-) = 12$ ;  $w_E(\pm) = 8$  for each member;  $\sum_\Gamma d_\Gamma w_\Gamma(\pm) = 32$

rule:  $\Gamma_{\text{space}} \otimes \Gamma_{\text{spin}}$  must contain the parity character,  $\chi_+ = A_1$  (solid) or  $\chi_- = A_2$  with  $b \mapsto -1$  (dashed);  $\chi_{\text{stat}} \equiv +1$  on  $G_6$ , every permutation there being even;  $A_2$  has no copy among the three version states (fig. 7) and is still a valid spin sector

level rules and connections in the manner of Harter (1993) fig. 4.2.3; a double rule marks the two-dimensional E